Skeletal Muscle Mitochondrial & Capillary Growth in Humans

Graphical Summary of Systematic Review & Meta-Regression (*Sports Medicine*)

> ***Meta-Analysis Scope:*** *Quantitative synthesis of* ***5,973 human participants*** *across* ***353 mitochondrial*** *and* ***131 capillary*** *exercise training studies.*
>
> *Mølmen, Knut Sindre, et al. "Effects of Exercise Training on Mitochondrial and Capillary Growth in Human Skeletal Muscle: A Systematic Review and Meta-Regression." Medicine [Auckland, N.Z.], vol. 55, no. 1, Jan. 2025, pp. 115–44. PubMed*
>
> ***Citekey****:: Molmen25trainMitochondCapill* ***ZoteroItemKey****:: K6GC6V5A*
>
> *Designed with AI advice: C:\Users\scott\obsidian\Main\lit\lit_notes\ai-searches\Molmen25trainMitochondCapill_IntensityVsResult_Graph.md*

Mitochondrial Expansion
23% – 27%
▲ Equal gain across ET, HIT, & SIT (P = 0.138)

Sprint Training Efficiency
3.9x – 4.9x
▲ SIT efficiency vs Endurance Training (P < 0.001)

Capillary Growth Peak
⩽ 4 Weeks
▲ Rapid angiogenesis (+13%), plateaus after 4w

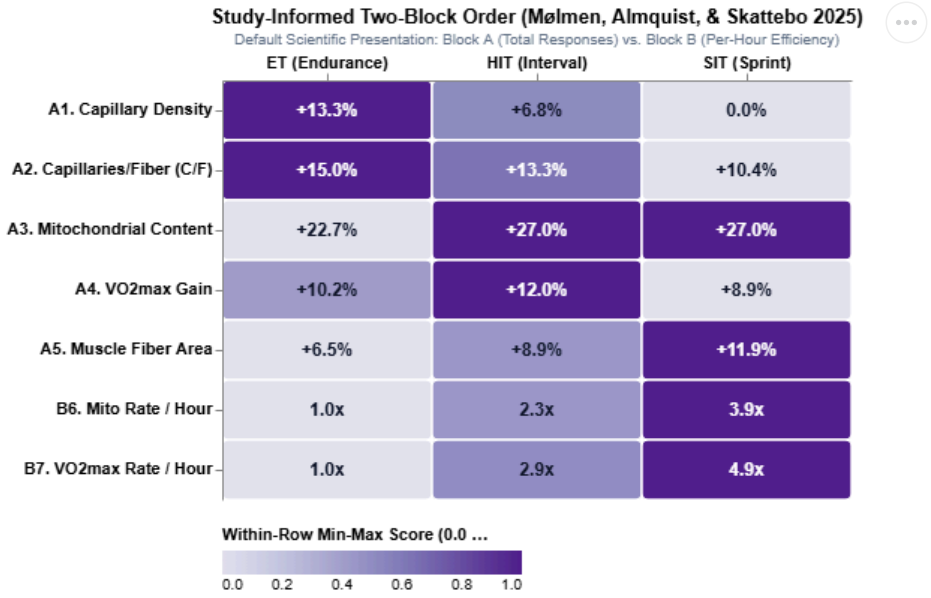
Lifelong Muscle Plasticity
100% Equal
▲ Equal adaptation across Age, Sex, &

Select Dashboard View:

- (●) 📋 Complete Visual Report (Spider + Bar Graphs + ALL)
- ( ) ⭐ Study-informed Two-Block Order (Default Scientific Presentation)
- ( ) 🧭 Heatmap Coherent Ordering Comparison (OLO vs. Gradient vs. Rank)
- ( ) 🔥 2x3 Heatmap Matrix Grid (Heatmaps 1–6)
- ( ) 🕸️ Spider / Radar Profile
- ( ) 📊 Exercise Intensity Category Bar Charts (ET vs HIT vs SIT)
- ( ) ⏱️ Time-Efficiency & Progression
- ( ) 👥 Population Adaptability (Age, Sex, Disease)
- ( ) 🧬 Physiological Pathways & Correlations

⭐ Study-Informed Two-Block Heatmap (Default Scientific Presentation)

Primary Source Citation: Mølmen, Almquist, and Skattebo (2025), Sports Medicine. All values represent **adjusted model estimates** (covariate-adjusted marginal means acr exercise intensity categories).



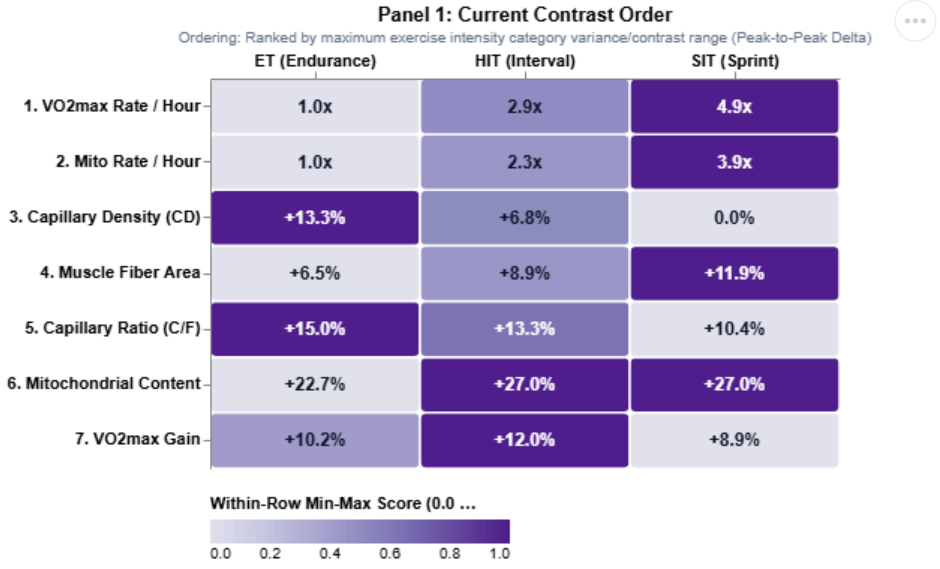
🔬 **Methodological & Physiological Notes (Mølmen, Almquist, & Skattebo 2025):**

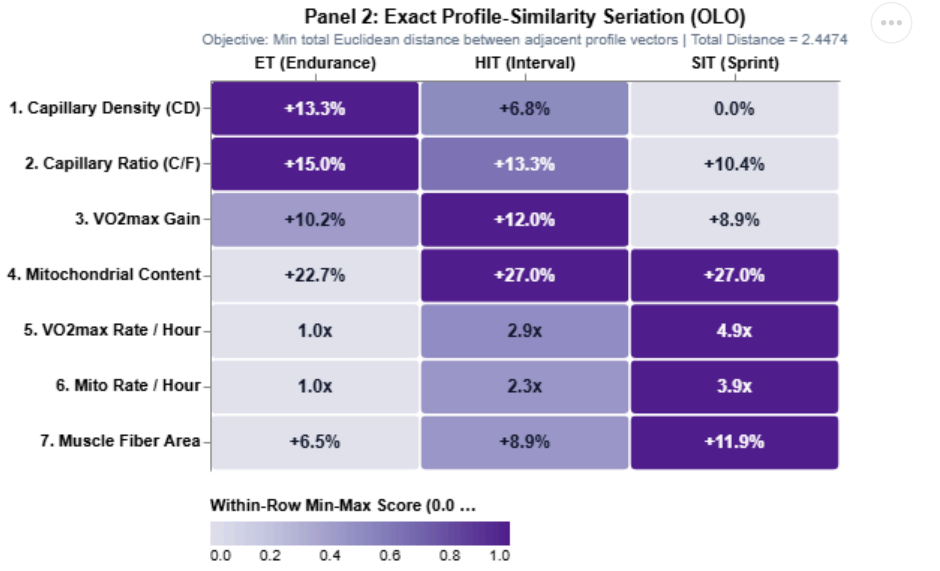
1. **Capillary Density vs. Capillaries per Fiber:** Capillary density is capillaries per mm^2, whereas capillaries per fiber (C/F) is a distinct morphological outcome; they shoul be described interchangeably.
2. **Hypertrophy Dilution Effect:** The higher ET capillary-density estimate (+13.3%) may partly reflect lower ET muscle-fiber hypertrophy (+6.5%), rather than a larger ca per-fiber response.
3. **Statistical Equivalence:** Mitochondrial-content (+23% to +27%) and capillaries-per-fiber changes (+10% to +15%) did not differ significantly among ET, HIT, and SIT (0.138) in the reported adjusted analyses.
4. **Efficiency vs. Total Response:** The final two rows (Block B) are per-active-exercise-hour efficiency estimates and should not be interpreted as directly comparable in absolute magnitude with the preceding percentage-change rows (Block A).
5. **Inference Guard:** Visual color differences represent exercise intensity category relative magnitude (0.0 to 1.0), not statistical significance. Original effect estimates a 95% confidence intervals remain the basis for inference.

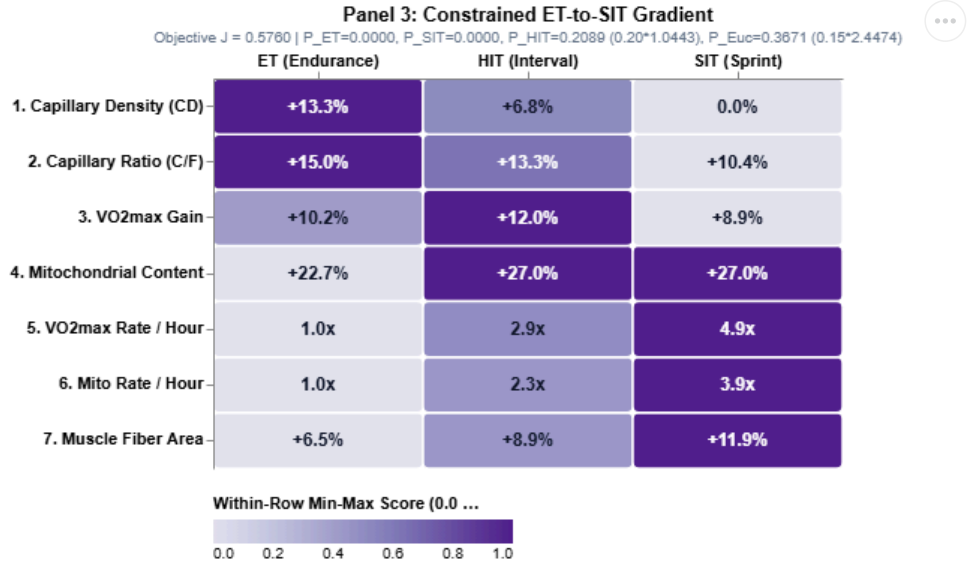
🧭 Heatmap Coherent Ordering Comparison (Exploratory Displays)

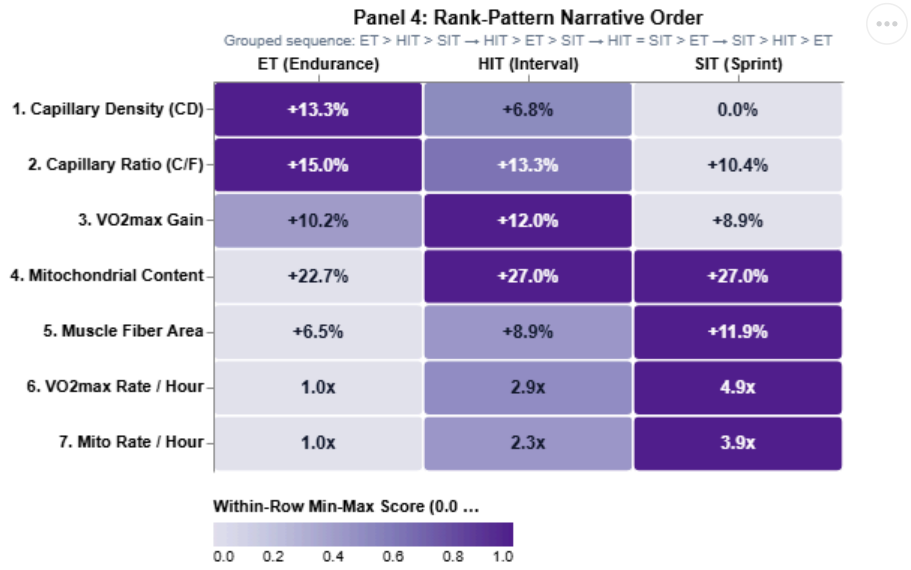
ℹ️ **Exploratory Ordering Note:**

Unconstrained OLO seriation, custom monotonic gradients, and rank patterns are offered as exploratory displays. However, the **Study-Informed Two-Block Order** is the scientific presentation because it preserves Mølmen et al. (2025)'s key distinction between total-training responses and per-hour efficiency.

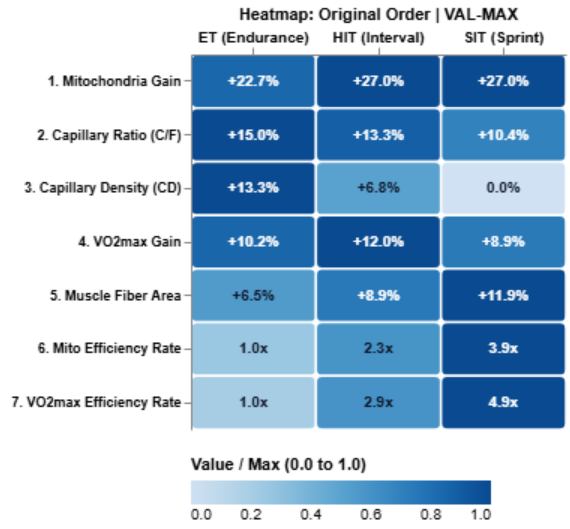


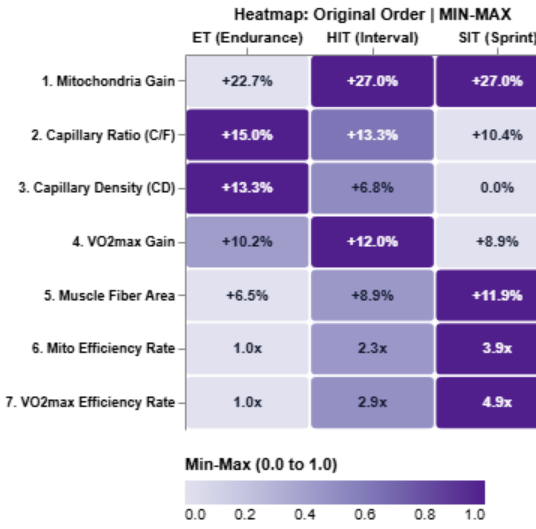


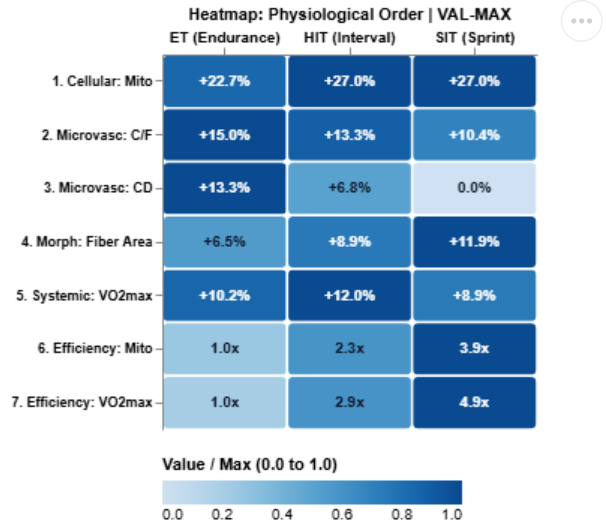


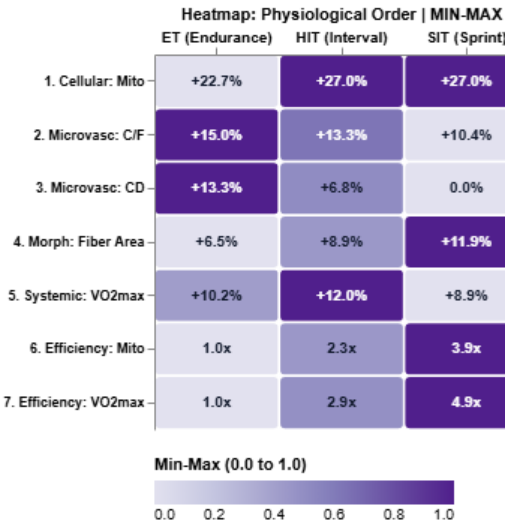


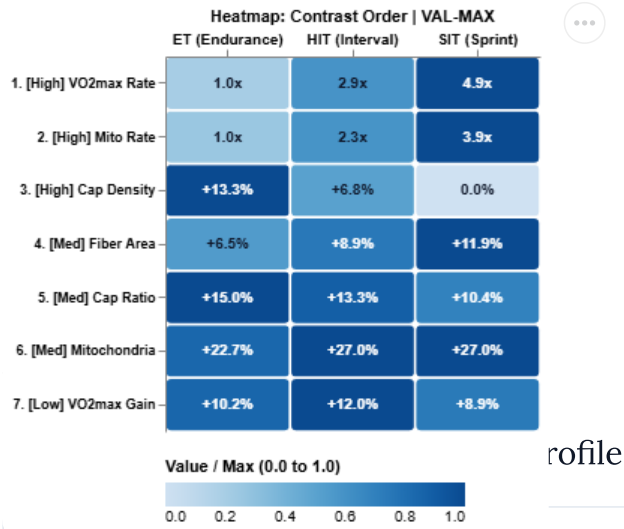
🔥 2x3 Heatmap Matrix: Ordering Strategies vs. Color Scales

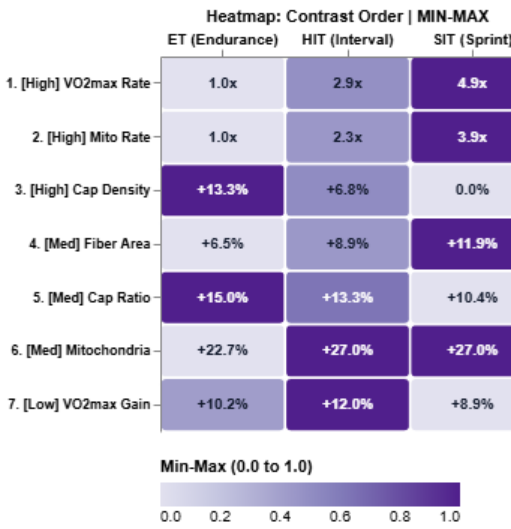




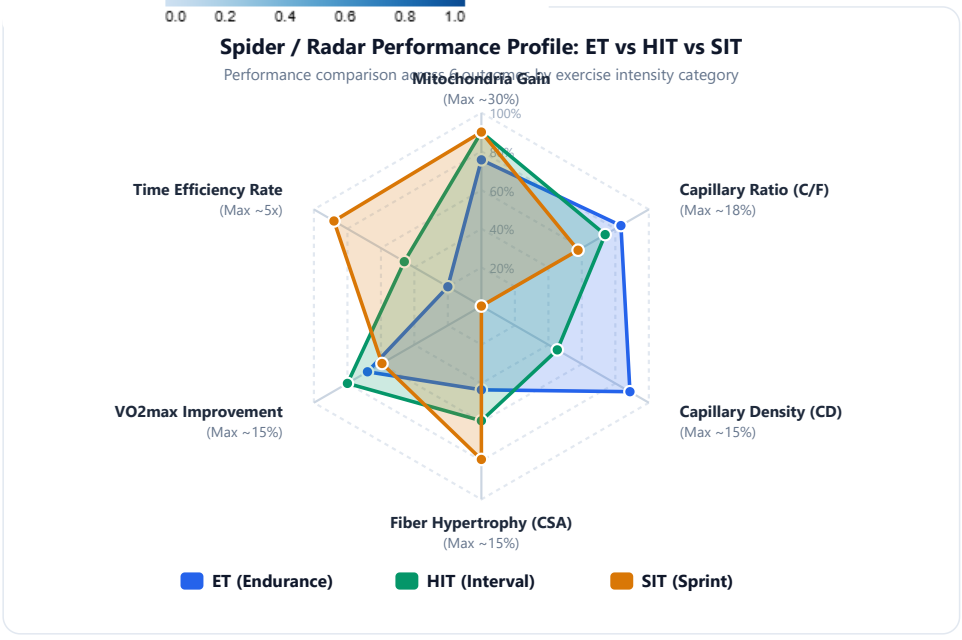

Heatmap: Physiological Order | VAL-MAX
ET (Endurance)
HIT (Interval)
SIT (Sprint)
1. Cellular: Mito
+22.7%
+27.0%
+27.0%
2. Microvasc: C/F
+15.0%
+13.3%
+10.4%
3. Microvasc: CD
+13.3%
+6.8%
0.0%
4. Morph: Fiber Area
+6.5%
+8.9%
+11.9%
5. Systemic: VO2max
+10.2%
+12.0%
+8.9%
6. Efficiency: Mito
1.0x
2.3x
3.9x
7. Efficiency: VO2max
1.0x
2.9x
4.9x
Value / Max (0.0 to 1.0)
0.0 0.2 0.4 0.6 0.8 1.0


Heatmap: Physiological Order | MIN-MAX
ET (Endurance)
HIT (Interval)
SIT (Sprint)
1. Cellular: Mito
+22.7%
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4. Morph: Fiber Area
+6.5%
+8.9%
+11.9%
5. Systemic: VO2max
+10.2%
+12.0%
+8.9%
6. Efficiency: Mito
1.0x
2.3x
3.9x
7. Efficiency: VO2max
1.0x
2.9x
4.9x
Min-Max (0.0 to 1.0)
0.0 0.2 0.4 0.6 0.8 1.0

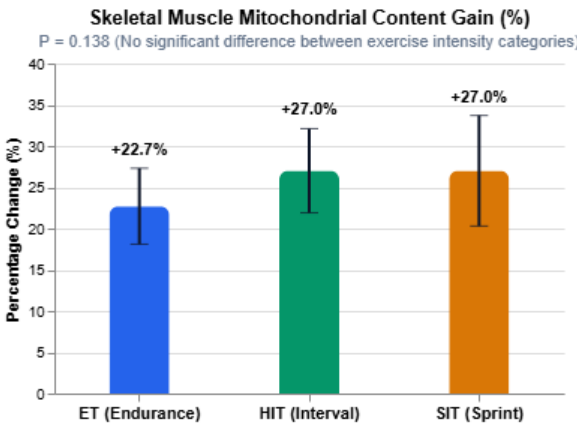

Heatmap: Contrast Order | VAL-MAX
ET (Endurance)
HIT (Interval)
SIT (Sprint)
1. [High] VO2max Rate
1.0x
2.9x
4.9x
2. [High] Mito Rate
1.0x
2.3x
3.9x
3. [High] Cap Density
+13.3%
+6.8%
0.0%
4. [Med] Fiber Area
+6.5%
+8.9%
+11.9%
5. [Med] Cap Ratio
+15.0%
+13.3%
+10.4%
6. [Med] Mitochondria
+22.7%
+27.0%
+27.0%
7. [Low] VO2max Gain
+10.2%
+12.0%
+8.9%
Value / Max (0.0 to 1.0)
0.0 0.2 0.4 0.6 0.8 1.0

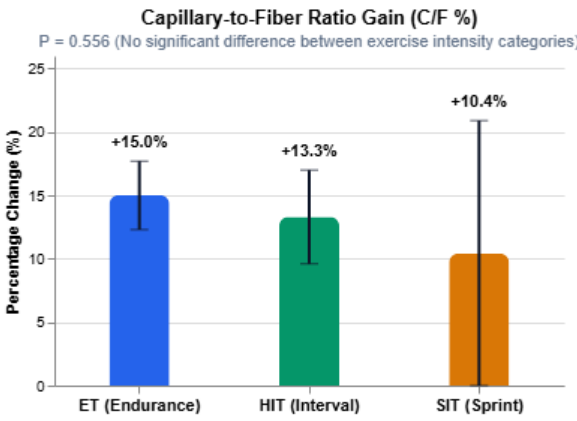

Heatmap: Contrast Order | MIN-MAX
ET (Endurance)
HIT (Interval)
SIT (Sprint)
1. [High] VO2max Rate
1.0x
2.9x
4.9x
2. [High] Mito Rate
1.0x
2.3x
3.9x
3. [High] Cap Density
+13.3%
+6.8%
0.0%
4. [Med] Fiber Area
+6.5%
+8.9%
+11.9%
5. [Med] Cap Ratio
+15.0%
+13.3%
+10.4%
6. [Med] Mitochondria
+22.7%
+27.0%
+27.0%
7. [Low] VO2max Gain
+10.2%
+12.0%
+8.9%
Min-Max (0.0 to 1.0)
0.0 0.2 0.4 0.6 0.8 1.0

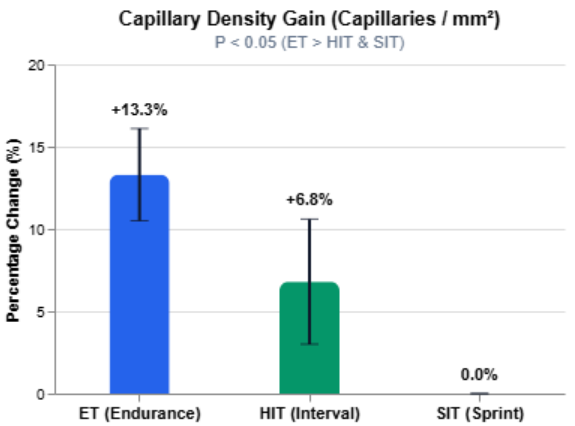
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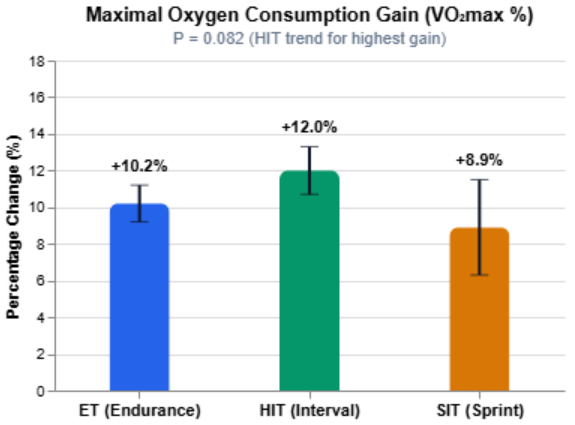

Spider / Radar Performance Profile: ET vs HIT vs SIT
Performance comparison across 6 adaptations by exercise intensity category
Mitochondria Gain
(Max ~30%)
100%
80%
60%
40%
20%
Capillary Ratio (C/F)
(Max ~18%)
Capillary Density (CD)
(Max ~15%)
Fiber Hypertrophy (CSA)
(Max ~15%)
VO2max Improvement
(Max ~15%)
Time Efficiency Rate
(Max ~5x)
ET (Endurance)
HIT (Interval)
SIT (Sprint)

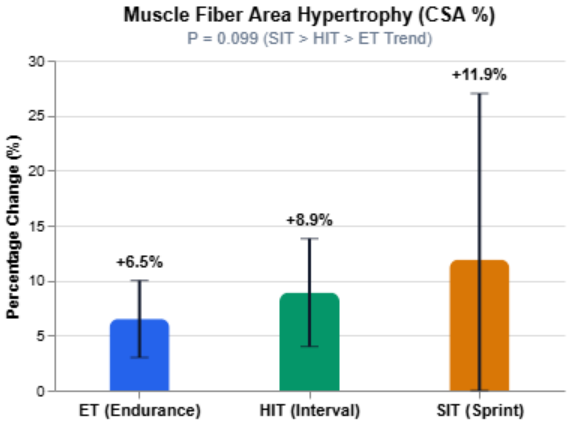
📊 4. Exercise Intensity Category Altair Bar Charts (ET vs HIT vs SIT)



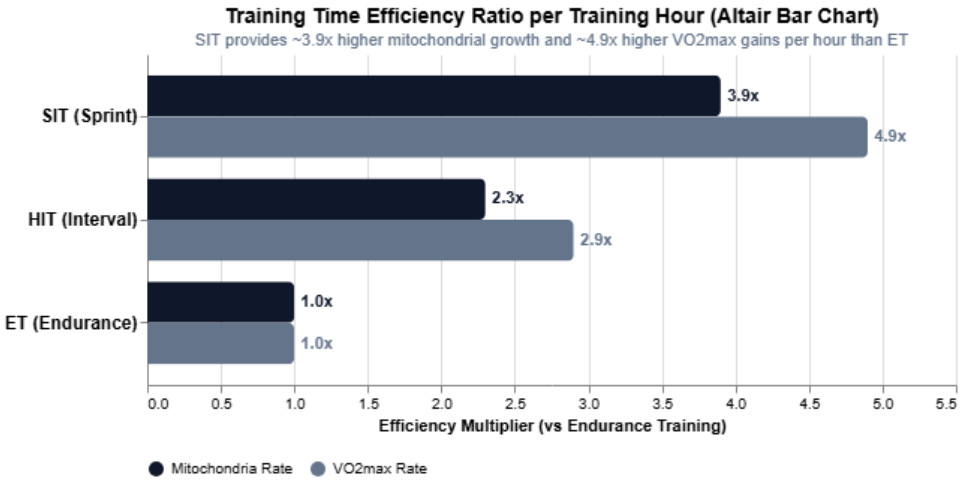


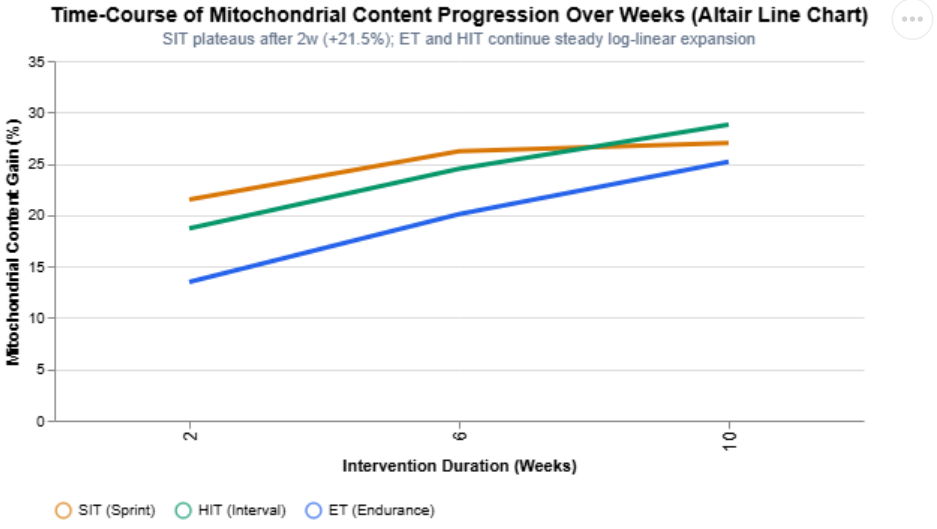






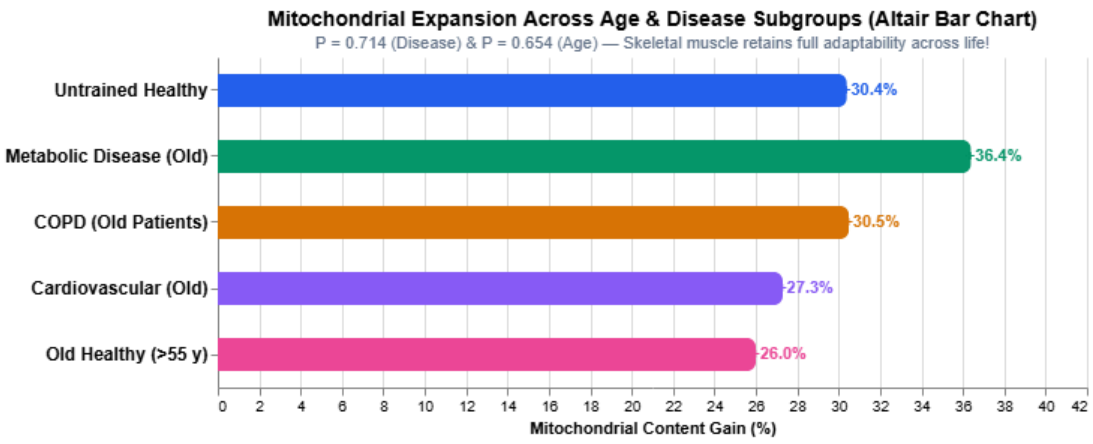
⏱️ 5. Training Time-Efficiency & Progression





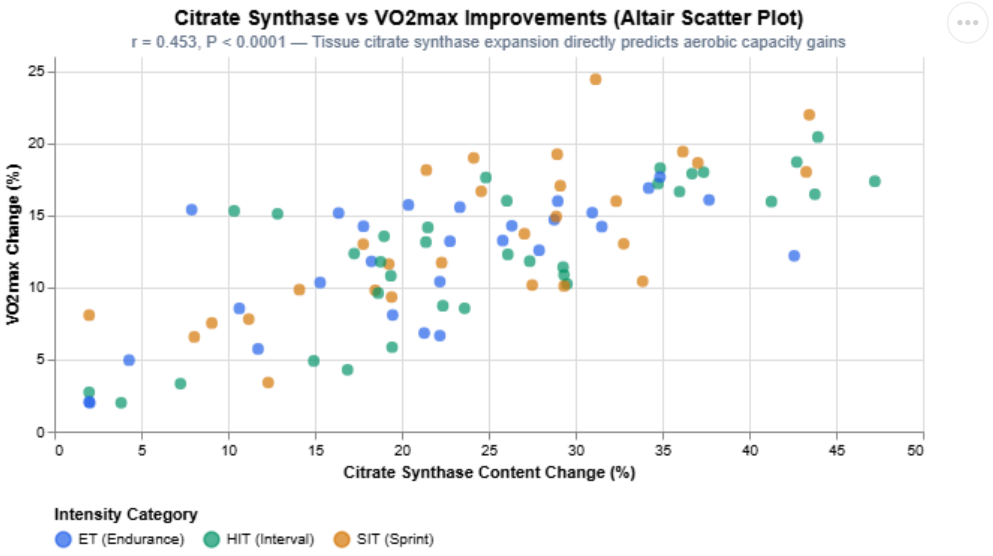
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6. Population Adaptability Matrix Across Subgroups



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7. Physiological Pathways & Meta-Regression Correlations



Practical Exercise Prescription Takeaways

1. **Equivalence of Exercise Intensity Categories:**
   - Low/moderate endurance training (ET), high-intensity interval training (HIT), and sprint interval training (SIT) all produce **equivalent percentage increases in skeletal muscle mitochondrial content (~23% to 27%)** and **capillary-to-fiber ratio (~10% to 15%)**.
2. **Time-Efficiency Advantage of SIT:**
   - Sprint Interval Training (SIT) requires **~8 hours total training time** compared to **~38.6 hours for ET**, making SIT **3.9x more time-efficient** for mitochondrial gro and **4.9x more efficient** for VO_2max gains per hour invested.
3. **Hypertrophy vs Capillary Density Trade-off:**
   - Higher exercise intensity (SIT/HIT) promotes greater muscle fiber hypertrophy (+11.9% CSA in SIT), which dilutes capillary density unless continuous endurar training (ET) is included to stimulate sustained capillary growth.
4. **Universal Adaptability Across Lifespan and Disease:**
   - Mitochondrial adaptation capacity remains **100% intact throughout human aging (>55 y)** and is **uncompromised by chronic metabolic, cardiovascular, or pulr diseases (COPD)**.
5. **Baseline Fitness is the Primary Modifier:**
   - Untrained individuals experience the largest relative gains (**+30.4% mitochondria, +15% C/F, +13.1% VO_2max**), whereas well-trained athletes show smaller marginal gains (**+6.6% mitochondria, +4.5% VO_2max**).